

David Duong

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EDUCATION

University of California, Irvine

Bachelor of Science in Computer Science - 3.92/4.00

Irvine, CA

December 2027

EXPERIENCE

First American

Jun. 2026 - Present

Data Engineer Intern

Santa Ana, CA

- Developing a full-stack data validation dashboard with React, FastAPI, and PostgreSQL to enable automated comparison of source and target datasets to support data migration efforts.
- Building a cross-platform data validation service using Apache Arrow, Parquet, and DuckDB to perform memory-efficient row-level comparisons, processing 24M+ row datasets in under 10 minutes.
- Engineering a near real-time data replication service from SQL Server to BigQuery using Google Pub/Sub, Dataflow, and Cloud Run for aggregating external API usage metrics.
- Implementing a reverse-ETL data synchronization pipeline from BigQuery to SQL Server using Google Pub/Sub and Dataflow to serve external users.

Temco Logistics

Mar. 2026 – Jun. 2026

Data Engineer (BI) Intern

Carson, CA

- Engineered a fuzzy-matching based data pipeline using Python and Spark in Databricks to migrate 50,000+ records from a legacy platform to cloud storage to serve internal legal, operations, and HR teams.
- Refactored a legacy ingestion service from pandas to Spark, reducing the codebase by over 50% with Azure Key Vault and Blob Storage integration.
- Optimized Spark transformation runtimes from over 2 hours to 30 minutes by utilizing multi-threading and partitioning strategies to maximize throughput.
- Designed and deployed test suites using PyTest and GitHub Actions to enforce CI/CD pipelines.

CareTech UCI

Oct. 2025 – Apr. 2026

Software Developer

Irvine, CA

- Developed a full-stack application with a team of 8 developers using React, FastAPI, and Supabase to track and visualize daily vitamin intake to promote cognitive health.
- Trained a YOLO26 computer vision model for meal classification by preprocessing image data through resizing, normalization, and augmentation, and iterating across model sizes to achieve 80%+ classification accuracy.

PROJECTS

Sentiment Data Pipeline | *Go, Python, SQL*

- Engineered services using Go and Python to ingest and classify 1,000+ data points per second.
- Optimized data ingestion by over 80% through the implementation of concurrent worker pools in Go.
- Implemented classification for 5,000+ datapoints by integrating a Hugging Face transformers model.

NASA Kibo Robot Programming Challenge | *Java, Python, PyTorch*

- Ranked 6th out of 55 teams nationally by engineering an Android application that autonomously commanded NASA's Astrobee robot to capture and classify images using NASA's APIs and OpenCV.
- Engineered a multithreaded Java pipeline to parallelize image capture and robot movement, maximizing efficiency.
- Built a data augmentation pipeline with Python, PyTorch, and Albumentations to synthetically generate training data to achieve 85%+ accuracy on a YOLOv5 model.

TECHNICAL SKILLS

Languages: Python, Java, Go, SQL, Typescript/Javascript, Bash

Technologies: Apache Spark, Apache Arrow, Delta Lake, Parquet, PostgreSQL, DuckDB, SQL Server, PyTorch

Frameworks: FastAPI, Node.js/Express.js, React

Cloud Platforms: Databricks, GCP (Pub/Sub, Cloud Run, Dataflow, BigQuery), Azure (Blob Storage, SQL Server, Key Vault)

Developer Tools: Git, Github Actions, Docker, Linux, Cursor, Codex